

Problem #1: Binary Evolution (36 points)

We have officially covered all the top level things necessary to understand the general principles of binary evolution! This problem asks you to prove it.

Problem #1a: Install and run COSMIC (12 pts)

COSMIC is a binary population synthesis code that can evolve binary stars from ZAMS up to the formation and merger of the stellar remnants in the binary. The first part of this problem is to install and import COSMIC into a jupyter notebook. Since COSMIC only builds for linux and mac os, if you have a windows machine, you should use Google Colab (which runs linux). You can follow the installation instructions in the [COSMIC documentation](#)

Once you have COSMIC installed, run the [first example](#) in the documentation which evolves a massive binary star that produces black holes which kind of match the mass of GW150914. To prove that you have run the code, print the 'tphys', 'mass_1', 'mass_2', 'kstar_1', 'kstar_2', 'porb', 'ecc', and 'evol_type' columns from the bpp DataFrame. You can use the 'utils.convert_kstar_evol_type' function in COSMIC by first importing utils, then calling the function with your bpp DataFrame as an argument as shown in the documentation. You can also read about the kstar and evol_type keys in the [COSMIC documentation](#).

Describe the evolution of the binary in words. Is the Roche lobe of either component in the binary filled? If so, how does the Roche overflow affect the orbit? What about the masses of each component in the binary? How does the formation of each black hole alter the orbit?

Problem #1b: Low mass binary evolution (12 pts)

Evolve a ZAMS binary with the following parameters:

$mass_1 = 6.0 M_{\odot}$, $mass_2 = 1.5 M_{\odot}$, $porb = 600$ days, $ecc = 0$, $Z = 0.014$

In the 'Evolve.evolve' function call, supply the keyword argument: 'dtp=0.0'. This will tell COSMIC to keep all timesteps for the binary in the 'bcm' DataFrame.

Once the binary is evolved, print the bpp DataFrame with the same columns as in Problem 1a. Next, make a figure of the orbital period vs time and a second figure of the two component masses vs time. Be sure to label each axis, and include a legend for the mass figure. If most of the evolution occurs at certain times, you should trim the x limits of your figure.

Similar to Problem 1a, describe the evolution of the binary in words taking care to describe the reason for all the different features in each of your figures.

Problem #1c: High mass binary evolution (12 pts)

Since natal kicks are implemented as a stochastic (random) process, if you want to reproduce the exact evolution of a binary, you'll need to supply a random seed to numpy. For this problem, you should seed numpy's random number generator with 5. Once the seed is set, evolve a ZAMS binary with the following parameters:

$mass_1 = 50.0 M_{\odot}$, $mass_2 = 35.0 M_{\odot}$, $porb = 120$ days, $ecc = 0$, $Z = 0.014$

In the 'Evolve.evolve' function call, supply the keyword argument: 'dtp=0.0'. As before, this will tell COSMIC to keep all timesteps for the binary in the 'bcm' DataFrame.

Once the binary is evolved, print the bpp DataFrame with the same columns as in Problem 1a. Next, make a figure of the orbital period vs time and a second figure of the two component masses vs time. Be sure to label each axis, and include a legend for the mass figure. If most of the evolution occurs at certain times, you should trim the x limits of your figure.

Similar to Problem 1a, describe the evolution of the binary in words taking care to describe the reason for all the different features in each of your figures. What is the main outcome difference from this binary and the binary in problem 1a?

Problem #2: Gravitational wave analysis (6 points)

The year is 2038. Prof. Breivik is as hype as she's ever been because the first data from LISA has been released. One signal has been measured to have a dimensionless strain value of $h = 1.8 \times 10^{-22}$, a gravitational-wave frequency of $f_{\text{GW}} = 2.75 \text{ mHz}$ and a gravitational-wave frequency evolution of $\dot{f}_{\text{GW}} = 1.1 \times 10^{-16} \text{ Hz/s}$.

Based on these measured quantities, solve for the chirp mass, $\mathcal{M}_c$, and distance, D to this new source. What type of source do you expect is producing this gravitational-wave signal based on its chirp mass?